Backdoor Defense through Self-Supervised and Generative Learning

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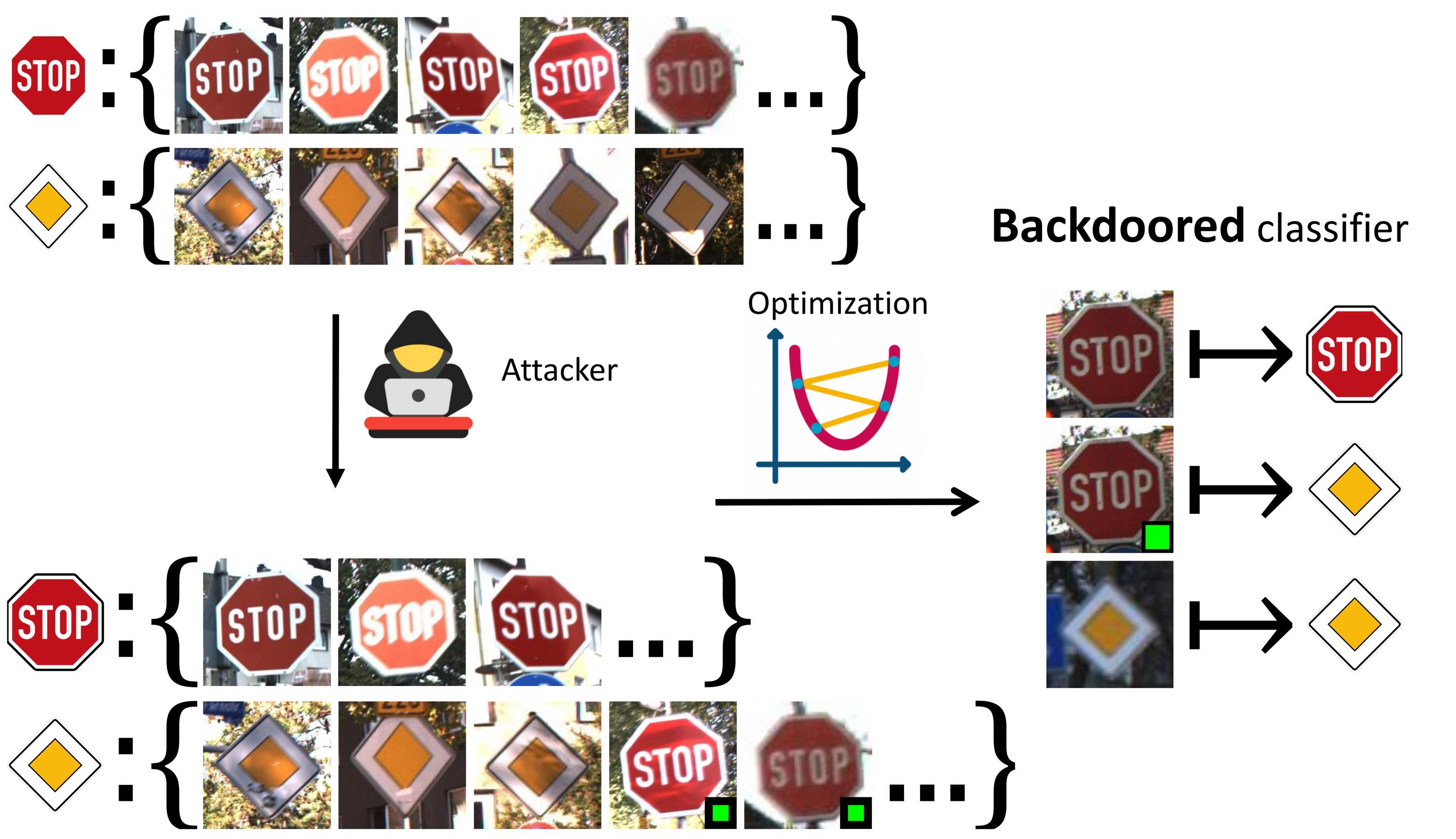
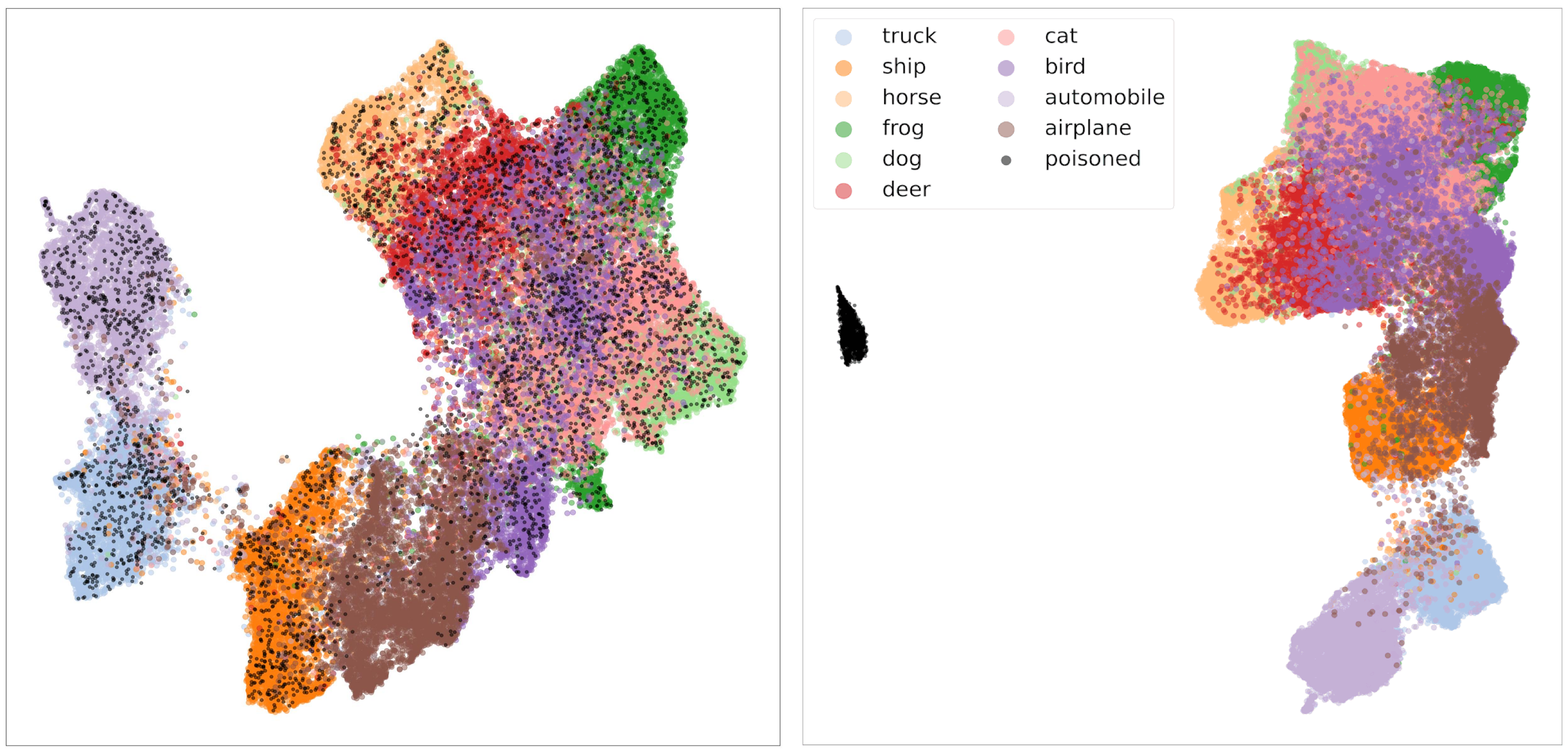
A CLASSIFIER ROBUST AGAINST BACKDOOR ATTACKS, *trained on data cleansed by generative modeling of self-supervised features*

Problem setup



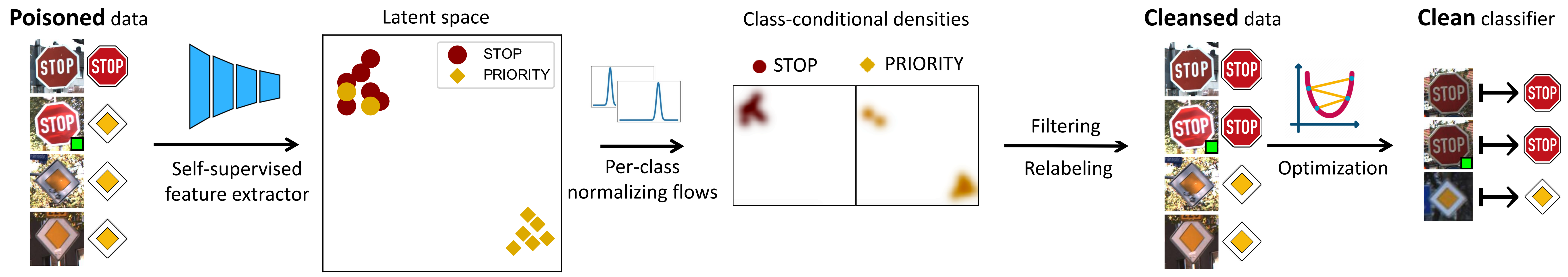
GOAL: Train a **robust** classifier given the **poisoned** dataset

Poisoned data in self-supervised latent space



Non-disruptive attacks | **Disruptive** attacks

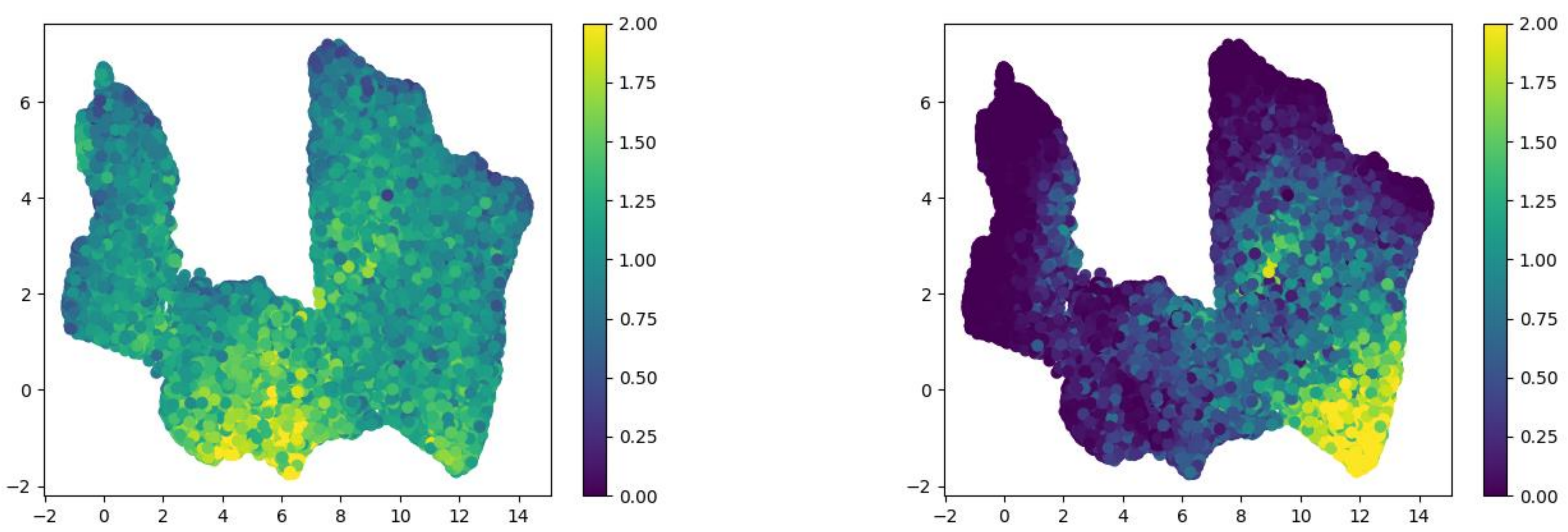
Generative modeling of self-supervised features



Filtering and relabeling

Target class detection

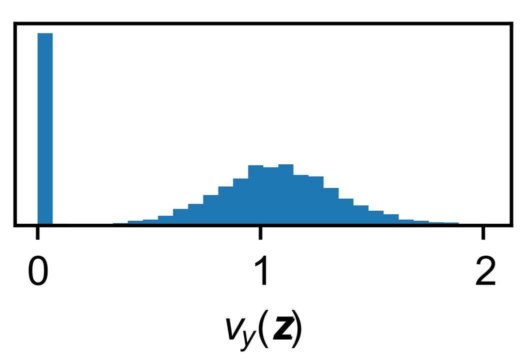
Non-disruptive attacks



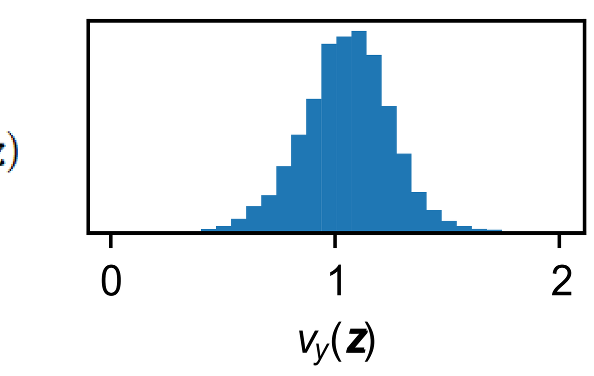
Density of inputs given **target** class is **HIGH**.

Density of inputs given **non-target** class is **LOW**.

Disruptive attacks



$$v_y(z) = \max_{y' \in \mathcal{Y}\setminus\{y\}} p_{\theta_{y'}}(z)$$

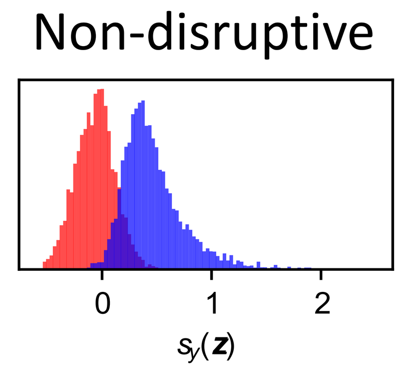


Target class contains **HIGH** number of examples with low density given other classes.

Non-target class contains **LOW** number of examples with low density given other classes.

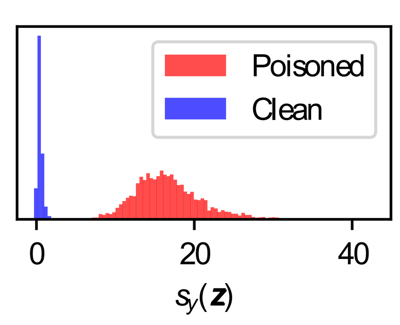
Dataset cleansing

Non-disruptive



$$s_y(z) = \frac{p_{\theta_y}(z)}{\max_{y' \in \mathcal{Y}\setminus\{y\}} p_{\theta_{y'}}(z)}.$$

Disruptive



Relabeling

$$\hat{\mathcal{D}}'_{\mathrm{P}} = \left\{ \left(\boldsymbol{x}, \arg\max_{y' \in \mathcal{Y}\setminus\{y\}} p_{\boldsymbol{\theta}_{y'}}(f_{\boldsymbol{\theta}_{\mathrm{F}}}(\boldsymbol{x})) \right) : (\boldsymbol{x}, y) \in \hat{\mathcal{D}}_{\mathrm{P}} \right\}$$

Experiments

- We showcase **robustness** against different families of attacks:

Dataset ↓	Defense →	No Defense		DBD		ASD		GSSD (ours)	
	Attack ↓	ACC	ASR	ACC	ASR	ACC	ASR	ACC	ASR
CIFAR-10	BadNets$_{.1}$	94.9	100.0	92.4	0.96	92.1	3.00	**91.7**	**0.14**
	Blend$_{.1}$	94.2	98.3	92.2	1.73	**93.4**	**1.00**	92.2	0.77
	WaNet$_{.1}$	94.3	98.0	91.2	0.39	93.3	1.20	**93.7**	**1.35**
	LC$_{.025}$	94.9	99.3	89.7	0.01	93.1	0.90	**92.8**	**0.06**
	ISSBA$_{.1}$	94.5	100.0	83.2	0.53	92.4	2.13	**93.9**	**0.62**
	Ada-Patch$_{.01}$	95.2	80.9	92.9	1.77	93.6	100	**92.4**	**0.23**
	Ada-Blend$_{.01}$	95.0	64.9	90.1	99.9	94.0	93.9	**92.7**	**0.22**
	Average	-	-	90.2	15.1	93.1	28.9	**92.6**	**0.48**
ImageNet-30	BadNets$_{.1}$	95.3	99.98	91.2	0.54	90.7	9.72	**94.8**	**0.00**
	Blend$_{.1}$	93.7	99.93	90.3	0.58	89.9	2.07	**93.5**	**0.45**
	WaNet$_{.1}$	93.5	100.0	90.5	0.48	88.8	2.89	**93.4**	**1.33**
	Average	-	-	90.7	0.53	89.8	4.89	**93.9**	**0.59**

The defense method with highest ACC − ASR is marked in **bold**.

- Our defense can leverage **CLIP** features:

Attack →	BadNets		Blend		WaNet	
Feature extractor ↓	ACC	ASR	ACC	ASR	ACC	ASR
RN-18 supervised	75.1	6.00	76.7	76.40	92.4	0.42
RN-50 CLIP	91.1	0.10	**93.0**	**0.88**	**94.0**	**0.81**
RN-18 self-sup (SimCLR)	**91.7**	**0.14**	92.2	0.77	93.7	1.35

Comparison of different feature extractors on CIFAR-10.

Conclusion & Limitations

- Self-supervised learning and generative modeling seem to be a promising pathway to more robust models.
- **Limitations:**
  - The entire defense is compromised if the target-class detection step fails, as seen in the case of an all-to-all attack.